Yihan Meng

AI Research Engineer · Generative models / Optimization / Industrial vision

Ph.D. candidate, Nanjing University · Research Intern, Guodian Nanjing Automation

First author: NeurIPS 2026 / PAKDD 2024 Oral

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Education

Nanjing University Ph.D. candidate, Computer Science and Technology — Sep 2023 — Present

Nanjing University B.S., Computer Science and Technology — Sep 2019 — Jun 2023

Honors & awards

National Scholarship for Graduate Students 2026 Outstanding Graduate Student 2025

Guodian Nanjing Automation Scholarship 2025 President's Scholarship 2023 People's Scholarship 2022

Selected research

TIDE Condition learning and trajectory control for diffusion models — **NeurIPS 2026** First author

Condition embedding optimization · Subspace projection · Adaptive guidance · DDIM inversion

TIDE: Trajectory-Aware Watermark Propagation for Text-to-Image Diffusion Models

Proposed a generative watermarking method combining condition learning with trajectory-aware control to address signal decay during diffusion denoising and interference between watermark guidance and text conditioning, improving robustness while preserving text semantics and generation quality.

- **Parameter-efficient watermark condition learning:** Injected a Fourier-domain watermark into an intermediate diffusion latent. Froze the pretrained VAE, U-Net, and text encoder while jointly learning the watermark spectrum and an auxiliary condition embedding. Combined short-horizon reconstruction, full-sampling content consistency, detection-margin, and perturbation-energy constraints to retain the signal through subsequent denoising.
- **Trajectory control through guidance decomposition:** Decomposed unconditional, text, and watermark noise predictions into separate guidance residuals. Estimated a local semantic-update subspace from past text-guidance directions and projected out conflicting watermark components. Adapted the watermark guidance budget to trajectory curvature, directional overlap, and residual strength to reduce interference with the original generation direction.
- **Inversion-based detection & cross-model evaluation:** Applied partial DDIM inversion to the final image to recover the injection-time latent for frequency-domain matching. Evaluated different prompt distributions and injection times on Stable Diffusion 1.5/2/2.1. Across seven attacks on SD 2.1, mean **TPR@1%FPR reached 98.6%**, exceeding ROBIN's 82.5% by **16.1 percentage points**; CLIP was 0.403, comparable to unwatermarked generation.

SASBO Joint optimization of discrete selection and continuous perturbation — **PAKDD 2024** First author · Oral

Stochastic binary optimization · ARM gradient estimation · ℓ_0 constraints · Binary-search pruning

SASBO: Sparse Attack via Stochastic Binary Optimization

Reformulated the ℓ_0 objective as stochastic binary optimization to address non-differentiable pixel selection and its coupling with perturbation magnitude.

- **Stochastic optimization:** Factorized perturbations into Bernoulli masks and continuous magnitudes. Alternated unbiased ARM gradient estimates for selection probabilities with gradient descent on magnitudes, jointly optimizing location and strength.
- **Sparsity-constrained solver:** Used the k-th largest selection probability as a dynamic threshold and binary search to prune redundant changes. Evaluated targeted and untargeted attacks on CIFAR-10 and ImageNet under magnitude constraints.

On targeted CIFAR-10 at $\varepsilon = 255$, achieved 100% success with mean ℓ_0 of 14.88 versus GreedyFool's 48.99—a 69.6% reduction.

Yihan Meng

Industry research & software engineering

Industry experience

Guodian Nanjing Automation Co., Ltd. Research Intern · Visual algorithm development — Jul 2025 — Present

IS-6000/VAS · Visual analysis for substation inspection

Completed Jun 2026

Researched industrial vision algorithms and developed the IS-6000/VAS system to address inconspicuous defects in complex substation equipment, false alarms under environmental changes, and limited cross-site generalization, supporting training, evaluation, and field use on domestic hardware and software.

- **Domain-prior-driven visual feature enhancement:** Addressed scarce subtle-defect samples and local anomalies obscured by equipment textures and backgrounds. Organized site context, equipment structure, and defect semantics into dynamic text conditions; used text encoding and domain adaptation to form priors for visual-feature fusion. Added multi-frame context to supplement and correct candidate-defect evidence, improving recognition stability under complex imaging conditions.
- **Multi-expert representation disentanglement & joint inference:** Built distinct representations for component semantics, defect discrimination, and operating states to reduce interference from equipment structure, background textures, and lighting changes. Combined cross-level feature checks, multi-task inference, and confidence fusion to assess candidate defects and reduce false alarms and missed detections in complex backgrounds.
- **Cross-site adaptation & engineering application:** Linked scene features to model capability profiles, connecting evaluation, staged deployment, field inference, and difficult-sample feedback to support model selection and iteration across sites. Piloted in Xinjiang, Shandong, and Liaoning; appraised by the Chinese Society for Electrical Engineering in June 2026, with the project's technical achievements rated **"internationally leading."**

Open-source engineering

EPUB Toolkit

Independent project 2026

Structured document processing engine · TypeScript / Python / Web Worker

Designed an EPUB 2/3 document processing engine for multi-volume restructuring and cross-platform delivery, supporting navigation reconstruction, resource deduplication, reversible structural splitting, and local browser execution.

- **Content restructuring & reversible recovery:** Built a unified content model from OPF Manifest, Spine, and NCX/Nav. Designed resource remapping and provenance metadata to resolve chapter ordering, navigation hierarchy, and path conflicts across volumes, enabling restoration of their original structure.
- **Resource processing & memory optimization:** Designed two-stage resource processing: deduplicated images and fonts by media type and SHA-256, then parsed and rewrote relative references in XHTML/CSS/SVG. Decompressed resources individually and used incremental writing and staged processing to reduce memory held by intermediate decoded objects.
- **Cross-runtime execution & consistency verification:** Shared a TypeScript core across browser and Node.js, managing asynchronous processing with a Web Worker, generation markers, and cancellation. Delivered a standard-library-only Python CLI and combined EPUBCheck, Vitest, and Playwright for format compliance, cross-language interoperability, and end-to-end regression verification.

github.com/yumizhi/lightweight_epub_merge_tool ↗ Live application ↗

Collaborative research

SAM Sparse perturbation optimization for cross-model transfer

IJCNN 2024 Third author

Cross-model meta-learning · Ensemble gradients · ℓ_0 projection · Black-box transfer

Sparse Attack with Meta-Learning

Combined cross-model meta-learning with projection-based sparsification to reduce source-model overfitting and improve transfer to unseen models under a limited pixel budget.

- **Cross-model tasks:** Sampled tasks from a surrogate-model pool, aggregated support-model gradients for multi-step updates, then refined perturbations through meta-testing on a held-out model. Repeated task sampling reduces reliance on any single model's gradient direction.
- **Constrained projection & sparsification:** After each update, projected the global perturbation onto joint pixel-count and magnitude constraints. Ranked pixels by reduction in projection error and retained the top k to preserve transfer information within the sparsity budget.

On untargeted CIFAR-10 at ε = 255 with a 50-pixel budget, mean ASR across six black-box models reached 85.5%, versus 68.4% for the single-model SA baseline.